

Apply Area Concepts to Solve Problems

Lesson 5-4

Name: Type your name **Date:** Today's date **Class:** Class period



TODAY'S OBJECTIVES

Content Objective: I can find the area of a composite figure by adding or subtracting the areas of basic shapes.

Language Objective: I can explain my steps using the words composite figure, decompose, add, and subtract.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Area of Composite Figures

Composite Figure

Decompose

Add

Subtract

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The space inside a figure made of basic shapes, found with $A = A_1 + A_2$, is the

2

A shape made of two or more basic shapes combined is a

3

To break a composite figure into basic shapes is to
it.

4

To find the total area of separate parts, I
their
areas.

5 To find the area of a shape with a piece removed, I
the missing part's area.

6 A math rule written with symbols is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How much will the carpet cost?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Area of Composite Figures** — The total space inside a figure made of basic shapes. Formula: $A = A_1 + A_2$ — add the parts (or subtract a missing piece).

Área de figuras compuestas — El espacio total dentro de una figura formada por figuras básicas. Fórmula: $A = A_1 + A_2$ — suma las partes (o resta la pieza que falta).

- ☐ **Composite Figure** — A shape made by putting two or more simple shapes together.

Figura compuesta — Una figura formada al juntar dos o más figuras simples.

- ☐ **Decompose** — To break a shape into smaller, simpler shapes.

Descomponer — Separar una figura en figuras más pequeñas y simples.

- ☐ **Add** — Add up the areas of the smaller shapes to get the total.

Sumar — Sumar las áreas de las figuras pequeñas para obtener el total.

- ☐ **Subtract** — Take away the area of a missing piece from a bigger shape.

Restar — Quitar el área de una parte que falta de una figura más grande.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The top area is ____ sq ft and the stem area is ____ sq ft. Total = ____ + ____ = ____ sq ft. Cost = ____ x \$5 = \$ ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We decompose the L-shape because we already know how to find a rectangle's area.

Evidence: Students explain that the L-shape is hard to measure directly, so breaking it into two rectangles lets them use $A = l \times w$ on each piece and add the results.

Reasoning: This evidence connects the definition of area of composite figures to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The top area is** ____ **sq ft and the stem area is** ____ **sq ft. Total =** ____ **+** ____ **=** ____ **sq ft.**
Cost = ____ **x \$5 = \$** ____.

- **My answer is** ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** ____.

Mi afirmación es ____.

- **I know because** ____.

Lo sé porque ____.

- **So** ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.